

neural network? Also, questions such as how many hidden layers should be present, what should the connection model between different layers be, etc, are complex and need considerable thought and experimentation. We will discuss more on the neural network architecture in subsequent columns. Now let us look at a few computer science interview questions:

1. Let's assume that you have been given five years' data of the daily stock prices of a company, and you have been asked to predict its stock prices with intervals of 30 days, for the next one year. How would you go about modelling this problem with a traditional machine learning based solution? What is the type of classifier you would use? What would be the input feature vector?
2. Consider the problem explained in Question (1). Instead of using a traditional machine learning solution, if you are asked to use a neural network based model, what type would you use?
3. In machine learning, what is the problem of generalisation? How is the learning model adapted to reduce the generalisation error as opposed to reducing the training or validation error? If a classifier has a high cross-validation accuracy, does it follow that its test accuracy would also be high? If yes, explain why. If not, explain the reason why you think it may not be so.
4. What is meant by regularisation in the supervised learning context?
5. Most of us are familiar with supervised learning using neural networks. What are the unsupervised learning methods in the context of neural networks?
6. You have been given the problem of sentiment analysis, where, given a sentence, you need to classify it as having a positive, negative or neutral sentiment. You have been given vast quantities of labelled data of example sentences with their target sentiment polarity annotated. Given that there can be words in the test input sentences which are not present in the training set examples, how would you approach the 'out of vocabulary' words in the test sentences? If you decide just to ignore them, what would be the impact on the test set classification accuracy? Can you devise a better approach to handling this issue?
7. In many of the data mining problems, statistical machine learning based approaches are considered as an alternative to traditional rule based approaches. This is all the more so in the case of natural language processing (NLP). For instance, consider the problem of named entity recognition (NER). While it is possible to use a rule based approach for NER, today, most of the NER techniques are based on statistical machine learning based approaches. In your opinion, what are the advantages and disadvantages of a rule based system versus a statistical machine learning based system for NER? Is it possible to combine the two approaches?
8. Traditionally, text inputs to statistical NLP systems have used one hot vector representation of words. For instance,

- given a vocabulary of size V , each word is represented by a vector of size V where only one index is set to 1 corresponding to that word, and all the remaining indices in the vector are set to zero. This is known as the one hot vector representation. However, of late, vector space word models have become quite popular as alternative input representations. Models such as word2vec (<https://radimrehurek.com/gensim/models/word2vec.html>) and Glove (<http://nlp.stanford.edu/projects/glove/>) are quite popular these days for input representation in statistical NLP systems, as opposed to representing words as one hot vector. What is the advantage of using vector space word models over traditional one hot vector representation? Are there any disadvantages with such vector space models?
9. In neural networks, traditionally, the neuron activation function has been the sigmoid function. Are there alternative activation functions that can be used? What is the advantage of using a sigmoid function as the activation function?
 10. What are the methods of reducing over-fitting in traditional machine learning systems as opposed to neural network based systems?

Now, let us look at a few pointers on the Web where you can find more information on neural networks and deep learning. There is a nice tutorial from Stanford, which covers both unsupervised feature learning and deep learning at http://ufldl.stanford.edu/wiki/index.php/UFLDL_Tutorial. Another good resource for beginners of deep learning is the blog of Christopher Olah: <http://colah.github.io/>. A formal introduction to deep learning and neural networks is available in the newly released book available at <http://www.deeplearningbook.org/>. For those who are interested in applying deep learning and neural networks for text, a Stanford course on deep learning for NLP taught by Richard Socher is available at <http://cs224d.stanford.edu/syllabus.html>. There is also a new course on 'Neural networks for machine learning' available through Coursera, taught by Prof. Geoffrey Hinton, which is very useful. An extensive reading list of deep learning related papers can be found at <http://deeplearning.net/reading-list/>.

If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at sandyasm_AT_yahoo_DOT_com. Till we meet again next month, happy programming!  

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